

1 Scattering Theory

satisfies the inequality¹

$$P_\beta \ln \left(\frac{P_\beta}{P_\alpha} \right) \geq P_\beta - P_\alpha.$$

The rate of change of the entropy is thus bounded by

$$-\frac{d}{dt} \int d\alpha P_\alpha \ln P_\alpha \geq \int d\alpha \int d\beta [P_\beta - P_\alpha] \frac{d\Gamma(\beta \rightarrow \alpha)}{d\alpha}$$

or interchanging variables of integration in the second term

$$-\frac{d}{dt} \int d\alpha P_\alpha \ln P_\alpha \geq \int d\alpha \int d\beta P_\beta \left[\frac{d\Gamma(\beta \rightarrow \alpha)}{d\alpha} - \frac{d\Gamma(\alpha \rightarrow \beta)}{d\beta} \right].$$

But the unitarity relation (3.6.18) tells us that the integral over α on the right-hand side vanishes, so we may conclude that the entropy always increases:

$$-\frac{d}{dt} \int d\alpha P_\alpha \ln P_\alpha \geq 0. \quad (3.6.20)$$

This is the ‘Boltzmann H -theorem.’ This theorem is often derived in statistical mechanics textbooks either by using the Born approximation, for which $|M_{\beta\alpha}|^2$ is symmetric in α and β so that $d\Gamma(\beta \rightarrow \alpha)/d\alpha = d\Gamma(\alpha \rightarrow \beta)/d\beta$, or by assuming time-reversal invariance, which would tell us that $|M_{\beta\alpha}|^2$ is unchanged if we interchange α and β and also reverse all momenta and spins. Of course, neither the Born approximation nor time-reversal invariance are exact, so it is a good thing that the unitarity result (3.6.18) is all we need in order to derive the H -theorem.

The increase of the entropy stops when the probability P_α becomes a function only of conserved quantities such as the total energy and charge. In this case the conservation laws require $d\Gamma(\beta \rightarrow \alpha)/d\alpha$ to vanish unless $P_\alpha = P_\beta$, so we can replace P_β with P_α in the first term of Eq. (3.6.19). Using Eq. (3.6.18) again then shows that in this case, P_α is time-independent. Here again, we need only the unitarity relation (3.6.18), not the Born approximation or time-reversal invariance.

1.1 Partial-Wave Expansions²

It is often convenient to work with the S -matrix in a basis of free-particle states in which all variables are discrete, except for the total momentum and energy. This is possible because the components of the momenta $\mathbf{p}_1, \dots, \mathbf{p}_n$ in an n -particle state of definite total momentum $\mathbf{p}$ and total energy E form a $(3n - 4)$ -dimensional compact space; for instance, for $n = 2$ particles in the center-of-mass frame with $\mathbf{p} = 0$, this space is a two-dimensional spherical surface. Any function on such a compact space may be expanded in a series of generalized

¹The difference between the left- and right-hand sides approaches the positive quantity $(P_\alpha - P_\beta)^2/2P_\beta$ for $P_\alpha \rightarrow P_\beta$, and has a derivative with respect to P_α that is positive- or negative-definite for all $P_\alpha > P_\beta$ or $P_\alpha < P_\beta$, respectively.

²This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

‘partial waves’, such as the spherical harmonics that are commonly used in representing functions on the two-sphere. We may thus define a basis for these n -particle states that apart from the continuous variables $\mathbf{p}$ and E is discrete: we label the free-particle states in such a basis as $|E, \mathbf{p}, N\rangle$, with the index N incorporating all spin and species labels as well as whatever indices are used to label the generalized partial waves. These states may conveniently be chosen to be normalized so that their scalar products are:

$$\langle E', \mathbf{p}', N' | E, \mathbf{p}, N \rangle = \delta(E' - E) \delta^3(\mathbf{p}' - \mathbf{p}) \delta_{N'N}. \quad (3.7.1)$$

The S -operator then has matrix elements in this basis of the form

$$\langle E', \mathbf{p}', N' | S | E, \mathbf{p}, N \rangle = \delta(E' - E) \delta^3(\mathbf{p}' - \mathbf{p}) S_{N'N}(E, \mathbf{p}), \quad (3.7.2)$$

where $S_{N'N}$ is a unitary matrix. Similarly, the T -operator, whose free-particle matrix elements $\langle \beta | T | \alpha \rangle$ are defined to be the quantities $T_{\beta\alpha}^+$ defined by Eq. (3.1.18), may be expressed in our new basis (in accordance with Eq. (3.4.12)) as

$$\langle E', \mathbf{p}', N' | T | E, \mathbf{p}, N \rangle = \delta^3(\mathbf{p}' - \mathbf{p}) M_{N'N}(E, \mathbf{p}) \quad (3.7.3)$$

and the relation (3.2.7) is now an ordinary matrix equation:

$$S_{N'N}(E, \mathbf{p}) = \delta_{N'N} - 2\pi i M_{N'N}(E, \mathbf{p}). \quad (3.7.4)$$

We shall use this general formalism in the following section; for the present we will concentrate on reactions in which the initial state involves just two particles.

For example, consider a state consisting of two non-identical particles of species n_1, n_2 with non-zero masses M_1, M_2 and arbitrary spins s_1, s_2 . In this case, the states may be labelled by their total momentum $\mathbf{p} = \mathbf{p}_1 + \mathbf{p}_2$, the energy E , the species labels n_1, n_2 , the spin z -components σ_1, σ_2 , and a pair of integers ℓ, m (with $|m| \leq \ell$) that specify the dependence of the state on the directions of, say, $\mathbf{p}_1$. Alternatively, we can form a convenient discrete basis by using Clebsch–Gordan coefficients [9] to combine the two spins to give a total spin s with z -component μ , and then using Clebsch–Gordan coefficients again to combine this with the orbital angular momentum ℓ with three-component m to form a total angular momentum j with three-component σ . This gives a basis of states $|E, \mathbf{p}, j, \sigma, \ell, s, n\rangle$ (with n a ‘channel index’ labelling the two particle species n_1, n_2), defined by their scalar products with the states of definite individual momenta and spin three-components:

$$\begin{aligned} \langle \mathbf{p}_1, \sigma_1; \mathbf{p}_2, \sigma_2; n' | E, \mathbf{p}, j, \sigma, \ell, s, n \rangle &= \left(\frac{|\mathbf{p}_1| E_1 E_2}{E} \right)^{-1/2} \delta^3(\mathbf{p} - \mathbf{p}_1 - \mathbf{p}_2) \\ &\times \delta \left(E - \sqrt{\mathbf{p}_1^2 + M_1^2} - \sqrt{\mathbf{p}_2^2 + M_2^2} \right) \delta_{n'n} \sum_{m, \mu} C_{s_1 s_2}(s, \mu; \sigma_1, \sigma_2) C_{\ell s}(j, \sigma; m, \mu) Y_\ell^m(\hat{\mathbf{p}}_1). \end{aligned} \quad (3.7.5)$$

where Y_ℓ^m are the usual spherical harmonics [24]. The factor $(|\mathbf{p}_1| E_1 E_2 / E)^{-1/2}$ is inserted so that in the center-of-mass frame these states will be properly normalized:

$$\begin{aligned} \langle E', \mathbf{p}', j', \sigma', \ell', s', n' | E, \mathbf{0}, j, \sigma, \ell, s, n \rangle \\ = \delta^3(\mathbf{p}') \delta(E' - E) \delta_{j'j} \delta_{\sigma'\sigma} \delta_{\ell'\ell} \delta_{s's} \delta_{n'n}. \end{aligned} \quad (3.7.6)$$

For identical particles to avoid double counting we must integrate only over half of the two-particle momentum space, so an extra factor $\sqrt{2}$ should appear in the scalar product (3.7.6).

In the center-of-mass frame the matrix elements of any momentum-conserving and rotationally-invariant operator O must take the form:

$$\langle E, \mathbf{p}', j', \sigma', \ell', s', n' | O | E, \mathbf{0}, j, \sigma, \ell, s, n \rangle = \delta^3(\mathbf{p}') O_{\ell' s' n', \ell s n}^j(E) \delta_{j' j} \delta_{\sigma' \sigma}. \quad (3.7.7)$$

(The fact that this is diagonal in j and σ follows from the commutation of O with $\mathbf{J}^2$ and J_3 , and the further fact that the coefficient of $\delta_{\sigma\sigma'}$ is independent of σ follows from the commutation of O with $J_1 \pm iJ_2$. This is a special case of a general result known as the Wigner–Eckart theorem [25].) Applying this to the operator M whose matrix elements are the quantities $M_{\beta\alpha}$, it follows that the scattering amplitude (3.6.7) in the center-of-mass system takes the form

$$\begin{aligned} f(\mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n \rightarrow \mathbf{k}'\sigma'_1, -\mathbf{k}'\sigma'_2, n') \\ \equiv -4\pi^2 \sqrt{\frac{k'E'_1 E'_2 E_1 E_2}{E^2 k}} M_{\mathbf{k}'\sigma'_1, -\mathbf{k}'\sigma'_2, n'; \mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n} \\ = -\frac{4\pi^2}{k} \sum_{j\sigma\ell m\ell' m' s\mu s' \mu'} C_{s_1 s_2}(s, \mu; \sigma_1, \sigma_2) C_{\ell s}(j, \sigma; m, \mu) \\ \times C_{s'_1 s'_2}(s', \mu'; \sigma'_1, \sigma'_2) C_{\ell' s'}(j, \sigma; m', \mu') Y_{\ell'}^{m'*}(\hat{\mathbf{k}}') Y_{\ell}^m(\hat{\mathbf{k}}) M_{\ell' s' n', \ell s n}^j(E). \end{aligned} \quad (3.7.8)$$

The differential scattering cross-section is $|f|^2$. We will take the direction of the initial momentum $\mathbf{k}$ to be along the three-direction, in which case

$$Y_{\ell}^m(\hat{\mathbf{k}}) = \delta_{m0} \sqrt{\frac{2\ell+1}{4\pi}}. \quad (3.7.9)$$

Integrating $|f|^2$ over the direction of the final momentum $\mathbf{k}'$ and respectively summing and averaging over final and initial spin three-components gives the total cross-section³ for transitions from channel n to n' :

$$\begin{aligned} \sigma(n \rightarrow n'; E) &= \frac{\pi}{k^2(2s_1+1)(2s_2+1)} \sum_{j\ell s\ell' s'} (2j+1) \\ &\times \left| \delta_{\ell' \ell} \delta_{s' s} \delta_{n' n} - S_{\ell' s' n', \ell s n}^j(E) \right|^2. \end{aligned} \quad (3.7.10)$$

³In deriving this result, we use standard sum rules [9] for the Clebsch–Gordan coefficients: first

$$\sum_{\sigma_1 \sigma_2} C_{s_1 s_2}(s, \mu; \sigma_1, \sigma_2) C_{s_1 s_2}(\bar{s}, \bar{\mu}; \sigma_1, \sigma_2) = \delta_{s\bar{s}} \delta_{\mu\bar{\mu}},$$

and the same with primes; then

$$\sum_{m\sigma} C_{\ell s}(j, \sigma; m, \sigma) C_{\ell' s}(\bar{j}, \bar{\sigma}; m, \bar{\sigma}) = \delta_{j\bar{j}} \delta_{\sigma\bar{\sigma}},$$

and finally

$$\sum_{\sigma\mu} C_{\ell s}(j, \sigma; 0, \mu) C_{\ell' s}(j, \sigma; 0, \mu) = \frac{2j+1}{2\ell+1} \delta_{\ell\ell'}.$$

Summing (3.7.10) over all two-body channels gives the total cross-section for all elastic or inelastic two-body reactions:

$$\sum_{n'} \sigma(n \rightarrow n'; E) = \frac{\pi}{k^2(2s_1 + 1)(2s_2 + 1)} \sum_{j\ell s} (2j + 1) \times [(\mathbf{1} - S^j(E))^\dagger (\mathbf{1} - S^j(E))]_{\ell sn, \ell sn}. \quad (3.7.11)$$

For comparison, Eqs. (3.7.8), (3.7.9), (3.7.4), and the Clebsch–Gordan sum rules give the spin-averaged forward scattering amplitude as

$$f(n; E) = \frac{i}{2k(2s_1 + 1)(2s_2 + 1)} \sum_{j\ell s} (2j + 1) [\mathbf{1} - S^j(E)]_{\ell sn, \ell sn}.$$

The optical theorem (3.6.10) then gives the total cross-section as

$$\sigma_{\text{total}}(n; E) = \frac{2\pi}{k^2(2s_1 + 1)(2s_2 + 1)} \sum_{j\ell s} (2j + 1) \text{Re}[\mathbf{1} - S^j(E)]_{\ell sn, \ell sn}. \quad (3.7.12)$$

If only two-body channels can be reached from the channel n at energy E , then the matrix $S^j(E)$ (or at least some submatrix that includes channel n) is unitary, and thus

$$[(\mathbf{1} - S^j(E))^\dagger (\mathbf{1} - S^j(E))]_{\ell sn, \ell sn} = 2 \text{Re}[\mathbf{1} - S^j(E)]_{\ell sn, \ell sn}, \quad (3.7.13)$$

so (3.7.12) and (3.7.11) are equal. On the other hand, if channels involving three or more particles are open, then the difference of (3.7.12) and (3.7.11) gives the total cross-section for producing extra particles:

$$\sigma_{\text{production}}(n; E) = \frac{\pi}{k^2(2s_1 + 1)(2s_2 + 1)} \sum_{j\ell s} (2j + 1) [\mathbf{1} - S^j(E)^\dagger S^j(E)]_{\ell sn, \ell sn}, \quad (3.7.14)$$

and this must be positive.

The partial wave expansion is particularly useful when applied to processes where the relevant part of the S -matrix is diagonal. This is the case, for instance, if the initial channel n contains just two spinless particles, and no other channels are open at this energy, as in $\pi^+ - \pi^+$ or $\pi^+ - \pi^0$ scattering at energies below the threshold for producing extra pions (provided one ignores weak and electromagnetic interactions). For a pair of spinless particles we have $j = \ell$, and angular momentum conservation keeps the S -matrix diagonal. It is also possible for the S -matrix to be diagonal in certain processes involving particles with spin; for instance in pion–nucleon scattering we can have $j = \ell + \frac{1}{2}$ or $j = \ell - \frac{1}{2}$, but for a given j these two states have opposite parity, so they cannot be connected by non-zero S -matrix elements. In any case, if for some n and E the S -matrix elements $S_{N', \ell sn}^j(E, \mathbf{0})$ all vanish unless N' is the two-body state j, ℓ, s, n , then unitarity requires that

$$S_{\ell' s' n', \ell sn}^j(E) = \exp[2i\delta_{j\ell sn}(E)] \delta_{\ell' \ell} \delta_{s' s} \delta_{n' n}, \quad (3.7.15)$$

where $\delta_{j\ell sn}(E)$ is a real phase, commonly known as the *phase shift*. This formula is also often used where the two-body part of the S -matrix is diagonal but channels containing

three or more particles are also open; in such cases the phase shift must have a positive imaginary part, to keep (3.7.14) positive. For real phase shifts, the elastic and total cross-sections are then given by Eq. (3.7.10) or Eq. (3.7.12) as:

$$\sigma(n \rightarrow n; E) = \sigma_{\text{total}}(n; E) = \frac{4\pi}{k^2(2s_1 + 1)(2s_2 + 1)} \sum_{j\ell s} (2j + 1) \sin^2 \delta_{j\ell sn}(E). \quad (3.7.16)$$

This familiar result is usually derived in non-relativistic quantum mechanics by studying the coordinate space wave function for a particle in a potential. The derivation given here is offered both to show that the partial wave expansion applies for elastic scattering even at relativistic velocities, and also to emphasize that it depends on no particular dynamical assumptions, only on unitarity and invariance principles.

It is also often useful to introduce phase shifts in dealing with problems where several channels are open, forming a few irreducible representations of some internal symmetry group. The classic example of such an internal symmetry is isotopic spin symmetry, for which the channel index n includes a specification of the isospins T_1, T_2 of the two particles together with their three-components t_1, t_2 ; the states in channel n may be expressed as linear combinations of the t th components of irreducible representations T , with coefficients given by the familiar Clebsch–Gordan coefficients $C_{T_1 T_2}(T, t; t_1, t_2)$. Suppose that for the channels and energy of interest the S -matrix is diagonal in ℓ and s as well as j, T , and t . Unitarity and isospin symmetry then allow us to write the S -matrix as

$$S_{\ell' s' T' t' \ell s T t}^j = \exp[2i\delta_{j\ell s T}(E)] \delta_{\ell' \ell} \delta_{s' s} \delta_{T' T} \delta_{t' t}, \quad (3.7.17)$$

with $\delta_{j\ell s T}(E)$ a real phase shift, t -independent according to the Wigner–Eckart theorem. The partial cross-sections can again be calculated from Eq. (3.7.10), and the total cross-section is given by Eq. (3.7.12) as

$$\begin{aligned} \sigma_{\text{total}}(t_1, t_2; E) &= \frac{4\pi}{k^2(2s_1 + 1)(2s_2 + 1)} \\ &\times \sum_{j\ell s T t} (2j + 1) C_{T_1 T_2}(T, t; t_1, t_2)^2 \sin^2 \delta_{j\ell s T}(E). \end{aligned} \quad (3.7.18)$$

For instance, in pion–pion scattering we have phase shifts $\delta_{\ell\ell 0 T}(E)$ with $T = 0$ or $T = 2$ for each even ℓ and $T = 1$ for each odd ℓ , while for pion–nucleon scattering we have phase shifts $\delta_{j, j \pm \frac{1}{2}, \frac{1}{2}, T}$ with $T = \frac{1}{2}$ or $T = \frac{3}{2}$.

We can gain some useful insight about the threshold behavior of the scattering amplitudes and phase shifts from considerations of analyticity that are nearly independent of any dynamical assumptions. Unless there are special circumstances that would produce singularities in momentum space, we would expect the matrix element $M_{\mathbf{k}'\sigma'_1, -\mathbf{k}'\sigma'_2, n'; \mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n}$ to be an analytic function⁴ of the three-momenta $\mathbf{k}$ and $\mathbf{k}'$ near $k = 0$ or $k' = 0$ or (for elastic

⁴For instance, in the Born approximation (3.2.8), M is proportional to the Fourier transform of the coordinate space matrix elements of the interaction, and hence is analytic at zero momenta as long as these matrix elements fall off sufficiently rapidly at large separations. The chief exception is for scattering involving long-distance forces, such as the Coulomb force.

scattering) $k = k' = 0$. Turning to the partial wave expansion (3.7.8) for M , we note that $k^\ell Y_\ell^m(\hat{\mathbf{k}})$ is a simple polynomial function of the three-vector $\mathbf{k}$, so in order for the matrix element to be an analytic function of the three-momenta $\mathbf{k}$ and $\mathbf{k}'$ near $k = 0$ or $k' = 0$, the coefficients $M_{\ell's'n',\ell sn}^j$ or equivalently $\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S_{\ell's'n',\ell sn}^j$ must go as $k^{\ell+1/2}k'^{\ell'+1/2}$ when k and/or k' go to zero. Hence for small k and/or k' , it is only the lowest partial waves in the initial and/or final state that contribute appreciably to the scattering amplitude. We have three possible cases:

Exothermic reactions

Here k' approaches a finite value as $k \rightarrow 0$, and in this limit $\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S_{\ell's'n',\ell sn}^j$ goes as $k^{\ell+1/2}$. The cross-section (3.7.11) in this case goes as $k^{2\ell-1}$, where ℓ is here the *lowest* orbital angular momentum that can lead to the reaction. In the most usual case $\ell = 0$, so the reaction cross-section goes as $1/k$. (This is the case, for instance, in the absorption of slow neutrons by complex nuclei, or for the annihilation of electron-positron pairs into photons at low energy, aside from the higher-order effects of Coulomb forces.) The reaction rate is the cross-section times the flux, which goes like k , so the rate for an exothermic reaction behaves like a constant for $k \rightarrow 0$. However, it is the cross-section rather than the reaction rate that determines the probability of absorption when a beam crosses a given thickness of target material, and the factor $1/k$ makes this probability very high for slow neutrons in an absorbing material like boron.

Endothermic Reactions

Here the reaction is forbidden until k reaches a finite threshold value, where $k' = 0$. Just above this threshold $\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S_{\ell's'n',\ell sn}^j$ goes as $(k')^{\ell'+1/2}$. The cross-section (3.7.11) in this case goes as $(k')^{2\ell'+1}$, where ℓ' is here the *lowest* orbital angular momentum that can be produced at threshold. In the most usual case $\ell' = 0$, so the reaction cross-section rises above threshold like k' , and hence like $\sqrt{E - E_{\text{threshold}}}$. (This is the case, for instance, in the associated production of strange particles, or the production of electron-positron pairs in the scattering of photons.)

Elastic Reactions

Here $k = k'$, so k and k' go to zero together. (This is the case where $n' = n$, or where n' consists of particles in the same isotopic spin multiplets as are those in n .) In elastic scattering the partial waves with $\ell = \ell' = 0$ are always present, so in the limit $k \rightarrow 0$ the scattering amplitude (3.7.8) becomes a constant:

$$\begin{aligned} f(\mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n \rightarrow \mathbf{k}'\sigma'_1, -\mathbf{k}'\sigma'_2, n') \\ \rightarrow \sum_{s\sigma} C_{s_1 s_2}(s, \sigma; \sigma_1, \sigma_2) C_{s'_1 s'_2}(s, \sigma; \sigma'_1, \sigma'_2) a_s(n \rightarrow n'), \end{aligned} \quad (3.7.19)$$

where a is a constant, known as the *scattering length*, defined by the limit

$$S_{0sn',0sn}^s \longrightarrow \delta_{n'n} + 2ika_s(n \rightarrow n') \quad (k = k' \rightarrow 0). \quad (3.7.20)$$

Summing $4\pi|f|^2$ over final spins and averaging over initial spins gives the total cross-section for the transition $n \rightarrow n'$ at $k = k' = 0$:

$$\sigma(n \rightarrow n'; k = 0) = \frac{4\pi}{(2s_1 + 1)(2s_2 + 1)} \sum_s (2s + 1) |a_s(n \rightarrow n')|^2. \quad (3.7.21)$$

The classic instance of the use of this formula is in neutron–proton scattering, where there are two scattering lengths, with the spin singlet length a_0 considerably larger than the spin triplet length a_1 .

The partial wave expansion can also be used to make a crude guess about the behavior of cross-sections at high energy. With decreasing wavelengths, we may expect scattering to be described more or less classically: a particle of momentum k and orbital angular momentum ℓ would have an impact parameter ℓ/k , and will therefore strike a disk of radius R if $\ell \leq kR$. This can be interpreted as a statement about S -matrix elements:

$$S_{\ell' s' n', \ell s n}^j \longrightarrow \begin{cases} 0, & \ell \ll kR_n, \\ 1, & \ell \gg kR_n, \end{cases} \quad (3.7.22)$$

where R_n is some sort of interaction radius for channel n . For a given $\ell \gg s$, there are $2s + 1$ values of j , all close enough to ℓ to approximate $2j + 1 \simeq 2\ell + 1$, so the sum over j and s in Eq. (3.7.12) merely gives a factor of order

$$\sum_{js} (2j + 1) = (2\ell + 1) \sum_s (2s + 1) = (2\ell + 1)(2s_1 + 1)(2s_2 + 1).$$

The total cross-section is then given for $k \gg 1/R_n$ by Eq. (3.7.12) as

$$\sigma_{\text{total}}(n; E) \longrightarrow \frac{2\pi}{k^2} \sum_{\ell \lesssim kR_n} (2\ell + 1) \longrightarrow 2\pi R_n^2. \quad (3.7.23)$$

In exactly the same way, Eq. (3.7.10) gives the elastic scattering cross-section

$$\sigma(n \rightarrow n; E) \longrightarrow \pi R_n^2. \quad (3.7.24)$$

The difference between Eqs. (3.7.23) and (3.7.24) gives an inelastic cross-section πR_n^2 , which is what we would expect for collisions with an opaque disk of radius R_n . (The somewhat surprising elastic scattering cross-section πR_n^2 may be attributed to diffraction by the disk.) On the other hand, if we assume along with Eq. (3.7.22) that $S_{\ell' s' n', \ell s n}^j$ is complex only for impact parameters ℓ/k within a small range of width $\Delta_n \ll R_n$ around $\ell/k = R_n$, then using the inequality $|\text{Im}(1 - S_{\ell' s' n', \ell s n}^j)| \leq 2$, the same analysis gives a bound on the real part of the forward scattering amplitude

$$|\text{Re } f(n; E)| \leq 2kR_n\Delta_n \ll |\text{Im } f(n; E)|. \quad (3.7.25)$$

The smallness of the real part of the forward scattering amplitude at high energy is confirmed by experiment.

So far we have not said anything about whether the interaction radius R_n itself may depend on energy. As a very crude guess, we may take R_n as the distance at which

the factor $\exp(-\mu r)$ in the Yukawa potential (1.2.74) takes a value proportional to some unknown power of E , in which case R_n goes as $\log E$ for $E \rightarrow \infty$, and the cross-sections go as $(\log E)^2$. As it happens, it has been rigorously shown [26] on the basis of very general assumptions that the total cross-section can grow no faster than $(\log E)^2$ for $E \rightarrow \infty$, and in fact the observed proton–proton total cross-section rises something like $(\log E)^2$ at high energies, so this rough picture of high-energy scattering does seem to have some correspondence with reality.

1.2 Resonances⁵

It often happens that the particles participating in a multi-particle collision can form an intermediate state consisting of a *single* unstable particle R , that eventually decays into the particles observed as the final state. If the total decay rate of R is small the cross-section exhibits a rapid variation (usually a peak), known as a *resonance*, at the energy of the intermediate state R .

We shall see that the behavior of the cross-sections near a resonance is pretty much prescribed by the unitarity condition alone, which is a good thing since there are a number of very different mechanisms that can produce a nearly stable state:

- (a) The simplest possibility is that the Hamiltonian can be decomposed into two terms, a ‘strong’ Hamiltonian $H_0 + H_s$, which has the particle R as an eigenstate, plus a weak perturbation H_w , which allows R to decay into various states, including the initial and final states α, β of our collision process. For instance there is a neutral particle, the Z^0 , with $j = 1$ and mass 91 GeV, that would be stable in the absence of the electroweak interactions. These interactions allow the Z^0 to decay into electron–positron pairs, muon–antimuon pairs, etc., but with a total decay rate that is much less than the Z^0 mass. In 1989 the Z^0 particle was seen as a resonance⁶ in electron–positron collisions at CERN and Stanford, in the reactions $e^+ + e^- \rightarrow Z^0 \rightarrow e^+ + e^-$, $e^+ + e^- \rightarrow Z^0 \rightarrow \mu^+ + \mu^-$, etc.
- (b) In some cases a particle is long-lived because there is a potential barrier that nearly prevents its constituents from escaping. The classic example is nuclear alpha decay: it may be energetically possible for a nucleus to emit an alpha particle (a He^4 nucleus) but the strong electrostatic repulsion between the alpha particle and the nucleus creates a barrier region around the daughter nucleus, which the alpha particle is classically forbidden to enter. The decay then can proceed only by quantum mechanical barrier penetration, and is exponentially slow. Such an unstable state shows up as a resonance in the scattering of the alpha particle on the daughter nucleus. For instance, the lowest-energy state of the Be^8 nucleus is unstable against decay into two alpha particles, and

⁵This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

⁶Incidentally, this example shows that a resonant state only needs to decay *relatively* slowly; the Z^0 lifetime is 2.6×10^{-25} seconds, which is not long enough for a Z^0 travelling near the speed of light to cross an atomic nucleus. What is important is that the decay rate is 36 times smaller than the rate $\hbar/M_Z$ of oscillation of the Z^0 wave function in its rest frame.

is seen as a resonance in $\text{He}^4\text{-He}^4$ scattering. (In addition to Coulomb barriers, there are also centrifugal barriers that help lengthen the life of alpha-, beta-, and gamma-unstable nuclei of high spin.)

- (c) It is possible for complicated systems to be nearly unstable for statistical reasons, without the presence of any potential barriers or weak interactions. For instance, an excited state of a heavy nucleus may be able to decay only if, through a statistical fluctuation, a large part of its energy is concentrated on a single neutron. This state will then show up as a resonance in the scattering of a neutron on the daughter nucleus.

These mechanisms for producing long-lived states are so different that it is truly fortunate that most of the properties of resonances follow from unitarity alone, without regard to the dynamical mechanism that produces the resonance.

First, let's consider the energy dependence of the matrix element for a reaction near a resonance. A wave packet $\int d\alpha g(\alpha)|\alpha; \text{in}\rangle \exp(-iE_\alpha t)$ of 'in' states has a time-dependence given by (3.1.19)

$$\int d\alpha g(\alpha)|\alpha; \text{in}\rangle e^{-iE_\alpha t} = \int d\alpha g(\alpha)|\alpha\rangle_0 e^{-iE_\alpha t} + \int d\beta |\beta\rangle_0 \int d\alpha \frac{e^{-iE_\alpha t} g(\alpha) T_{\beta\alpha}^+}{E_\alpha - E_\beta + i\epsilon}.$$

As mentioned in Section 3.1, a pole in the function $T_{\beta\alpha}^+$ in the lower-half complex E_α plane would make a contribution to the second term that decays exponentially as $t \rightarrow \infty$. Specifically, a pole at $E_\alpha = E_R - i\Gamma/2$ yields a term in the *amplitude* that behaves like $\exp(-iE_R t - \Gamma t/2)$, so it corresponds to a state whose *probability* decays like $\exp(-\Gamma t)$. We conclude then that a long-lived state of energy E_R with a slow decay rate Γ produces a term in the scattering amplitude that varies as

$$T_{\beta\alpha}^+ \sim (E_\alpha - E_R + i\Gamma/2)^{-1} + \text{constant}. \quad (3.8.1)$$

To go further, it will be convenient to adopt as a basis the orthonormal discrete multi-particle states $|\mathbf{p}, E, N\rangle$ discussed in the previous section; $\mathbf{p}$ and E are the total momentum and energy, and N is an index that takes only discrete (though infinitely many) values. In this basis, the S -matrix may be written

$$S_{\mathbf{p}'E'N', \mathbf{p}EN} = \delta^3(\mathbf{p}' - \mathbf{p})\delta(E' - E)S_{N'N}(\mathbf{p}, E). \quad (3.8.2)$$

Near a resonance, we expect the center-of-mass frame amplitude $S(\mathbf{0}, E) \equiv \mathcal{S}(E)$ to have the form

$$\mathcal{S}_{N'N}(E) \equiv S_{N'N}(\mathbf{0}, E) = \mathcal{S}_{0N'N} + \frac{\mathcal{R}_{N'N}}{E - E_R + i\Gamma/2}, \quad (3.8.3)$$

where $\mathcal{S}_0$ and $\mathcal{R}$ are approximately constant at least over the relatively small range of energies $|E - E_R| \lesssim \Gamma$.

In this basis, the unitarity of the S -matrix is an ordinary matrix equation

$$\mathcal{S}(E)^\dagger \mathcal{S}(E) = \mathbf{1}. \quad (3.8.4)$$

Applied to Eq. (3.8.3), this tells us that the non-resonant background S -matrix is unitary

$$\mathcal{S}_0^\dagger \mathcal{S}_0 = \mathbf{1}, \quad (3.8.5)$$

and also that the residue matrix $\mathcal{R}$ satisfies the two conditions

$$\mathcal{S}_0^\dagger \mathcal{R} - \mathcal{R}^\dagger \mathcal{S}_0 = 0, \quad (3.8.6)$$

$$-\frac{i}{2}\Gamma \mathcal{S}_0^\dagger \mathcal{R} + \frac{i}{2}\Gamma \mathcal{R}^\dagger \mathcal{S}_0 + \mathcal{R}^\dagger \mathcal{R} = 0. \quad (3.8.7)$$

These conditions can be put in a more transparent form by setting

$$\mathcal{R} \equiv -i\Gamma \mathcal{A} \mathcal{S}_0. \quad (3.8.8)$$

The unitarity conditions on the matrix $\mathcal{A}$ are then simply

$$\mathcal{A}^\dagger = \mathcal{A}, \quad \mathcal{A}^2 = \mathcal{A}. \quad (3.8.9)$$

Any such Hermitian idempotent matrix is called a *projection matrix*. Such matrices can always be expressed as a sum of dyads of orthonormal vectors $u^{(r)}$:

$$\mathcal{A}_{N'N} = \sum_r u_{N'}^{(r)} u_N^{(r)*}, \quad \sum_N u_N^{(r)*} u_N^{(s)} = \delta_{rs}. \quad (3.8.10)$$

The discrete part of the S -matrix is then

$$\mathcal{S}_{N'N}(E) = \sum_{N''} \left[\delta_{N'N''} - i \frac{\Gamma}{E - E_R + i\Gamma/2} \sum_r u_{N'}^{(r)} u_{N''}^{(r)*} \right] \mathcal{S}_{0N''N}. \quad (3.8.11)$$

Each term in the sum over r can be thought of as arising from a different resonant state, all these states having the same values for E_R and Γ .

What has this to do with rates and cross-sections? For simplicity let's now ignore the non-resonant background scattering, setting $\mathcal{S}_{0N'N}$ equal to $\delta_{N'N}$; we will come back to the more general case a little later. Then for the two-body discrete center-of-mass states described in the previous section, Eq. (3.8.11) reads:

$$\begin{aligned} \mathcal{S}_{j'\sigma'\ell's'n',j\sigma\ell sn}(E) &= \delta_{j'j} \delta_{\sigma'\sigma} \delta_{\ell'\ell} \delta_{s's} \delta_{n'n} \\ &\quad - i \frac{\Gamma}{E - E_R + i\Gamma/2} \sum_r u_{j'\sigma'\ell's'n'}^{(r)} u_{j\sigma\ell sn}^{(r)*}. \end{aligned} \quad (3.8.12)$$

In all cases the label r will include an index σ_R giving the z -component of the total angular momentum of the resonant state; for a resonant state of total angular momentum j_R , σ_R takes $2j_R + 1$ values. If there is no other degeneracy, then r just labels the value of σ_R , and

$$u_{j\sigma\ell sn}^{(\sigma_R)} = \delta_{jj_R} \delta_{\sigma\sigma_R} u_{\ell sn}, \quad (3.8.13)$$

where $u_{\ell sn}$ are a set of complex amplitudes that (because of the Wigner–Eckart theorem) are independent of σ . Now Eq. (3.8.12) gives the amplitude S^j defined by Eq. (3.7.7) as

$$S_{\ell's'n',\ell sn}^j(E) = \delta_{\ell'\ell} \delta_{s's} \delta_{n'n} - i \delta_{jj_R} \frac{\Gamma}{E - E_R + i\Gamma/2} u_{\ell's'n'}^* u_{\ell sn}. \quad (3.8.14)$$

Also, Eq. (3.8.10) now reads

$$\sum_{\ell sn} |u_{\ell sn}|^2 + \dots = 1 \quad (3.8.15)$$

with the dots representing the positive contribution of any states containing three or more particles. As we shall see, the quantities $|u_{\ell sn}|^2$ have the interpretation of branching ratios for the decay of the resonant state into the various accessible two-body states.

Eq. (3.7.12) now gives the total cross-section for all reactions in channel n :

$$\sigma_{\text{total}}(n; E) = \frac{\pi(2j_R + 1)}{k^2(2s_1 + 1)(2s_2 + 1)} \frac{\Gamma \Gamma_n}{(E - E_R)^2 + \Gamma^2/4}, \quad (3.8.16)$$

where

$$\Gamma_n \equiv \Gamma \sum_{\ell s} |u_{\ell sn}|^2. \quad (3.8.17)$$

This is a version of the celebrated Breit–Wigner single-level formula [27]. We can also use these results to calculate the cross-section for resonant scattering from an initial two-body channel n to a final two-body channel n' . Using Eq. (3.8.14) in Eq. (3.7.10) gives

$$\sigma(n \rightarrow n'; E) = \frac{\pi(2j_R + 1)}{k^2(2s_1 + 1)(2s_2 + 1)} \frac{\Gamma_n \Gamma_{n'}}{(E - E_R)^2 + \Gamma^2/4}. \quad (3.8.18)$$

This shows that the probabilities that the resonant state will decay into any one of the final two-body channels n' are proportional to the $\Gamma_{n'}$. According to Eq. (3.8.15), the sum of the Γ_n (including contributions from final states containing three or more particles) is just equal to the total decay rate Γ , so we can conclude that Γ_n is just the rate for the decay of the resonant state into channel n .

We see in Eqs. (3.8.16) and (3.8.18) the characteristic resonant peak at energy E_R , with a width (the full width at half maximum) equal to the decay rate Γ . (The individual Γ_n are often called partial widths.) Since $\Gamma_n \leq \Gamma$, the total cross-section at the peak of the resonance is roughly bounded by one square wavelength, $(2\pi/k)^2$. This rule, that cross-sections at a single resonance are roughly bounded by a square wavelength, is universally applicable even in classical physics (where energy conservation plays the role played here by unitarity), as for instance in the resonant interaction of sound waves with bubbles in the sea, or gravitational waves with gravitational wave antennae. (In the latter case, the branching ratio for oscillations in any laboratory mass to lose their energy through gravitational radiation is tiny, so the cross-section even at a resonance peak is vastly less than a square wavelength [28].)

Incidentally, it often happens that a resonance is detected, but energy measurements are insufficiently precise to resolve its width. In this case, what is measured experimentally is the integral of the cross-section over the resonant peak. For the total cross-section (3.8.14), this is

$$\int \sigma_{\text{total}}(n; E) dE = \frac{2\pi^2(2j_R + 1)\Gamma_n}{k_R^2(2s_1 + 1)(2s_2 + 1)}. \quad (3.8.19)$$

Such experiments can reveal only the partial width for decay of the resonant state into the initial particles, not the total width or branching ratios.

This formalism can also be applied when the resonant states with a given spin z -component form a multiplet related by some symmetry group. For instance, to the extent that isospin symmetry is respected, for a resonance of total isospin T_R the index r labelling resonant states includes a specification not only of the angular-momentum z -component σ_R , but also of an isospin three-component t_R , taking values $-T_R, -T_R + 1, \dots, T_R$. In this case there is no change in the above results for the total and partial cross-sections, because each two-body channel n has definite values t_1, t_2 of the isospin z -components of the two particles, and hence can only couple to the resonant state with the single value $t_1 + t_2$ for t_R . The partial widths Γ_n here depend on t_1 and t_2 only through factors $C_{T_1 T_2}(T_R, t_R; t_1, t_2)^2$.

The presence of a resonance shows itself in a characteristic behavior of phase shifts near the resonance. Returning to the general formula (3.8.11) (but still taking $\mathcal{S}_0 = \mathbf{1}$), we see from Eq. (3.8.10) that for each individual resonant state r , there is an eigenvector $u_N^{(r)}$ of $\mathcal{S}_{N'N}(E)$ with eigenvalue

$$\exp[2i\delta^{(r)}(E)] = 1 - i \frac{\Gamma}{E - E_R + i\Gamma/2}$$

or, in other words,

$$\tan \delta^{(r)}(E) = - \frac{\Gamma/2}{E - E_R}. \quad (3.8.20)$$

We see that over an energy range of order Γ centered around the resonant energy, the the ‘eigenphase’ $\delta^{(r)}(E)$ jumps from a value $\nu\pi$ (with ν a positive or negative integer) below the resonance to $(\nu + 1)\pi$ above it. However, in order to use this result to say something about reaction rates, we need to know the eigenvectors $u_N^{(r)}$, which, in general, have components with arbitrary numbers of particles with various momenta, spin, and species.

These results are much more useful in those special cases when the particles in a particular channel N are forbidden (usually by conservation laws) from making a transition to any other channel. With this assumption, it is not difficult to include the effects of a non-resonant background scattering matrix $\mathcal{S}_0$ in the general result (3.8.11). In order for $\mathcal{S}_{N'N}$ to vanish for some particular N and all $N' \neq N$, it is necessary that the same is true of $\mathcal{S}_{0N'N}$, and also true of $u_{N'}^{(r)}$ for any r for which $u_N^{(r)} \neq 0$. The unitarity requirement (3.8.5) then requires that for this N

$$\mathcal{S}_{0N'N} = \exp(2i\delta_{0N})\delta_{N'N}$$

and Eq. (3.8.10) requires that

$$u_N^{(r)*} u_N^{(s)} = \delta_{rs},$$

so that there can be only one term r in Eq. (3.8.11) for which $u_N^{(r)} \neq 0$. In this case, Eq. (3.8.11) gives

$$\mathcal{S}_{N'N}(E) = \delta_{N'N} \left[1 - \frac{i\Gamma}{E - E_R + i\Gamma/2} \right] \exp(2i\delta_{0N}) \equiv \delta_{N'N} \exp[2i\delta_N(E)]$$

with total phase shift

$$\delta_N(E) = \delta_{0N} - \arctan\left(\frac{\Gamma/2}{E - E_R}\right). \quad (3.8.21)$$

We see that over an energy range of order Γ centered around the resonant energy E_R , the phase shift $\delta_N(E)$ jumps from a value δ_{0N} below the

resonance to $\delta_{0N} + \pi$ above it. For instance, as we saw in the previous section, these assumptions are satisfied in various two-body reactions such as pion–pion and pion–nucleon scattering at energies below the threshold for producing extra pions, with N incorporating the total and orbital angular-momenta j, ℓ (with $j = \ell$ for pion–pion scattering) and the total angular-momentum z -component σ , as well as the total isospin T and its three-component t . The Wigner–Eckart theorem allows the phase shifts to depend only on j, ℓ , and T , not on t or σ . There are famous resonances in these channels: in pion–pion scattering there is a resonance at 770 MeV called ρ with $j = \ell = 1$, $T = 1$, and $\Gamma = 150$ MeV; in pion–nucleon scattering there is a resonance at 1232 MeV called Δ with $j = \frac{3}{2}$, $\ell = 1$, $T = \frac{3}{2}$ and $\Gamma = 110$ to 120 MeV.

Inspection of Eq. (3.7.12) or Eq. (3.7.18) shows that the total cross-section reaches a peak when the resonant phase shift passes through $\pi/2$ (or odd-integer multiples of $\pi/2$.) The non-resonant phase shifts are typically rather small, so as we saw earlier, σ_{total} will exhibit a sharp peak when the phase shift δ_ℓ goes through $\pi/2$, at an energy close to E_R . However, it sometimes happens that the non-resonant background phase shift δ_{0N} is near $\pi/2$, in which case the cross-section will exhibit a sharp *dip* as the phase shift rises through π near E_R , due to destructive interference between the resonance and the non-resonant background amplitude. Such dips were first observed by Ramsauer and Townsend [29] in 1922, in the scattering of electrons by noble gas atoms.

Problems

1. Consider a theory with a separable interaction; that is,

$$\langle \beta | H_{\text{int}} | \alpha \rangle = g u_\beta u_\alpha^*,$$

where g is a real coupling constant, and u_α is a set of complex quantities with

$$\sum_\alpha |u_\alpha|^2 = 1.$$

Use the Lippmann–Schwinger equation (3.1.16) to find explicit solutions for the ‘in’ and ‘out’ states and the S -matrix.

2. Suppose that a resonance of spin one is discovered in e^+e^- scattering at a total energy of 150 GeV and with a cross-section (in the center-of-mass frame, averaged over initial spins, and summed over final spins) for elastic e^+e^- scattering at the peak of the resonance equal to 10^{-34} cm^2 . What is the branching ratio for the decay of the resonant state R by the mode $R \rightarrow e^- + e^+$? What is the total cross-section for e^+e^- scattering at the peak of the resonance? (In answering both questions, ignore the non-resonant background scattering.)
3. Express the differential cross-section for two-body scattering in the *laboratory* frame, in which one of the two particles is initially at rest, in terms of kinematic variables

and the matrix element $M_{\beta\alpha}$. (Derive the result directly, without using the results derived in this chapter for the differential cross-section in the center-of-mass frame.)

4. Derive the perturbation expansion (3.5.8) directly from the expansion (3.5.3) of old-fashioned perturbation theory.
5. We can define ‘standing wave’ states $|\alpha\rangle^0$ by a modified version of the Lippmann–Schwinger equation

$$|\alpha\rangle^0 = |\alpha\rangle_0 + \text{P.V.} \frac{1}{E_\alpha - H_0} H_{\text{int}} |\alpha\rangle^0.$$

Show that the matrix $K_{\beta\alpha} = \pi\delta(E_\beta - E_\alpha)\langle\beta|H_{\text{int}}|\alpha\rangle^0$ is Hermitian. Show how to express the S -matrix in terms of the K -matrix.

6. Express the differential cross-section for elastic π^+ –proton and π^- –proton scattering in terms of the phase shifts for states of definite total angular momentum, parity, and isospin.
7. Show that the states $|E, \mathbf{p}, j, \sigma, s, n\rangle$ defined by Eq. (3.7.5) are correctly normalized to have the scalar products (3.7.6).

References

1. For more details, see M. L. Goldberger and K. M. Watson, *Collision Theory* (John Wiley & Sons, New York, 1964); R. G. Newton, *Scattering Theory of Waves and Particles*, 2nd edn (Springer-Verlag, New York, 1982).
- 1a. B. Lippmann and J. Schwinger, *Phys. Rev.* **79**, 469 (1950).
2. J. A. Wheeler, *Phys. Rev.* **52**, 1107 (1937); W. Heisenberg, *Z. Phys.* **120**, 513, 673 (1943).
3. M. Born, *Z. Phys.* **37**, 863 (1926); **38**, 803 (1926).
4. C. Møller, *Kgl. Danske Videnskab. Mat. Fys. Medd.* **23**, No. 1 (1945); **22**, No. 19 (1946).
5. G. D. Rochester and C. C. Butler, *Nature* **160**, 855 (1947). For a historical review, see G. D. Rochester, in *Pions to Quarks – Particle Physics in the 1950s*, ed. by L. M. Brown, M. Dresden, and L. Hoddeson (Cambridge University Press, Cambridge, UK, 1989).
6. G. Breit, E. U. Condon, and R. S. Present, *Phys. Rev.* **50**, 825 (1936); B. Cassen and E. U. Condon, *Phys. Rev.* **50**, 846 (1936); G. Breit and E. Feenberg, *Phys. Rev.* **50**, 850 (1936).
7. M. A. Tuve, N. Heydenberg, and L. R. Hafstad, *Phys. Rev.* **50**, 806 (1936).

8. M. Gell-Mann, Cal. Tech. Synchrotron Laboratory Report CTSL-20 (1961); *Phys. Rev.* **125**, 1067 (1962); Y. Ne'eman, *Nucl. Phys.* **26**, 222 (1961).
9. See e.g. A. R. Edmonds, *Angular Momentum in Quantum Mechanics* (Princeton University Press, Princeton, 1957): Chapter 3 (where $C_{j_1 j_2}(jm; m_1 m_2)$ is denoted $(j_1 j_2 jm | j_1 m_1 j_2 m_2)$); M. E. Rose, *Elementary Theory of Angular Momentum* (John Wiley & Sons, New York, 1957): Chapter III (where $C_{j_1 j_2}(jm; m_1 m_2)$ is denoted $C(j_1 j_2 j; m_1 m_2 m)$).
10. G. Feinberg and S. Weinberg, *Nuovo Cimento Serie X*, **14**, 571 (1959).
11. W. Chinowsky and J. Steinberger, *Phys. Rev.* **95**, 1561 (1954); also see B. Ferretti, *Report of an International Conference on Fundamental Particles and Low Temperatures*, Cambridge, 1946 (The Physical Society, London, 1947).
12. T. D. Lee and C. N. Yang, *Phys. Rev.* **104**, 254 (1956).
13. C. S. Wu *et al.*, *Phys. Rev.* **105**, 1413 (1957).
14. R. Garwin, L. Lederman, and M. Weinrich, *Phys. Rev.* **105**, 1415 (1957); J. I. Friedman and V. L. Telegdi, *Phys. Rev.* **105**, 1681 (1957).
15. K. M. Watson, *Phys. Rev.* **88**, 1163 (1952).
16. M. Gell-Mann and A. Pais, *Phys. Rev.* **97**, 1387 (1955); also see A. Pais and O. Piccioni, *Phys. Rev.* **100**, 1487 (1955).
17. J. H. Christenson, J. W. Cronin, V. L. Fitch, and R. Turlay, *Phys. Rev. Letters* **13**, 138 (1964).
18. K. R. Schubert *et al.*, *Phys. Lett.* **31B**, 662 (1970). This reference analyzes neutral kaon data without the assumption of CPT invariance, and finds that the part of the CP-violating amplitude that conserves CPT and violates T has both real and imaginary parts five standard deviations from zero, while the part that conserves T and violates CPT is within a standard deviation of zero.
19. R. H. Dalitz, *Phil. Mag.* **44**, 1068 (1953); also see E. Fabri, *Nuovo Cimento* **11**, 479 (1954).
20. F. J. Dyson, *Phys. Rev.* **75**, 486, 1736 (1949).
21. See, e.g., L. I. Schiff, *Quantum Mechanics*, 1st edn (McGraw-Hill, New York, 1949): Section 19.
22. This was first proved in classical electrodynamics. See, e.g., H. A. Kramers, *Atti Congr. Intern. Fisici, Como*, 1927; reprinted in H. A. Kramers, *Collected Scientific Papers* (North-Holland, Amsterdam, 1956). For the proof in quantum mechanics, see E. Feenberg, *Phys. Rev.* **40**, 40 (1932); N. Bohr, R. E. Peierls, and G. Placzek, *Nature* **144**, 200 (1939).

23. A general version of this argument was given in the late 1960s in unpublished work of C. N. Yang and C. P. Yang. Also see A. Aharony, in *Modern Developments in Thermodynamics* (Wiley, New York, 1973): pp. 95–114, and references therein.
24. See, e.g., A. R. Edmonds, *Angular Momentum in Quantum Mechanics* (Princeton University Press, Princeton, 1957): Chapter 2; M. E. Rose, *Elementary Theory of Angular Momentum* (John Wiley & Sons, New York, 1957): Appendix III; L. D. Landau and E. M. Lifshitz, *Quantum Mechanics – Non Relativistic Theory*, 3rd edn (Pergamon Press, Oxford, 1977): Section 28.
25. E. P. Wigner, *Gruppentheorie* (Friedrich Vieweg und Sohn, Braunschweig, 1931); C. Eckart, *Rev. Mod. Phys.* **2**, 305 (1930).
26. M. Froissart, *Phys. Rev.* **123**, 1053 (1961).
27. G. Breit and E. P. Wigner, *Phys. Rev.* **49**, 519 (1936).
28. See, e.g., S. Weinberg, *Gravitation and Cosmology* (Wiley, New York, 1972): Section 10.7.
29. R. Kollath, *Phys. Zeit.* **31**, 985 (1931).